DSA210 Project Proposal

Analysis of Bitcoin's Relationship with Traditional Financial Assets

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1. Introduction

Over the past decade, Bitcoin (BTC) has emerged as a significant financial asset, attracting attention from investors, researchers, and policymakers. It has been characterized both as a high-risk speculative asset and as a potential store of value comparable to gold. However, there is still ongoing debate regarding its true role within financial markets.

This project aims to analyze Bitcoin's behaviour in relation to traditional financial assets, specifically the S&P 500 index and Gold. By applying data science techniques, the study seeks to determine whether Bitcoin behaves more like a risk asset (similar to equities) or a safe-haven asset (similar to gold).

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2. Objectives

The main objectives of this project are:

- To analyze the relationship between Bitcoin, the S&P 500, and Gold
- To measure the correlation between these assets over time
- To examine whether Bitcoin behaves more like a risk asset or a safe-haven asset
- To identify how these relationships change under different market conditions

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3. Data Collection

The data for this project will be obtained from publicly available financial sources. Historical daily price data will be collected using Python-based financial APIs, specifically the Yahoo Finance API via the yfinance library.

The datasets will include:

- Bitcoin price data (BTC-USD)
- S&P 500 index data (^GSPC)
- Gold price data (GC=F or XAU/USD)

The data will cover a multi-year time period to ensure sufficient observations for meaningful analysis.

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4. Data Enrichment and Preparation

To enhance the quality and depth of the analysis, the collected data will be enriched through additional processing steps. These include:

- Calculating daily returns and percentage changes
- Aligning datasets by date to ensure consistency across assets
- Handling missing values and ensuring data cleanliness
- Computing rolling statistics such as moving averages and rolling correlations

These transformations will allow for a more detailed understanding of how relationships between assets evolve over time.

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5. Methodology

The project will follow a structured data science workflow:

5.1 Exploratory Data Analysis (EDA)

- Visualization of price trends over time
- Analysis of return distributions
- Identification of patterns and anomalies in the data

5.2 Statistical Analysis

- Calculation of correlation coefficients between Bitcoin, the S&P 500, and Gold
- Analysis of rolling correlations to observe how relationships change dynamically

5.3 Extended Analysis (Planned)

- Examination of asset behavior during major market events
- Comparison of Bitcoin's performance in different economic conditions
- Application of basic machine learning techniques if applicable

All analysis will be conducted using Python to ensure reproducibility and consistency.

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6. Expected Outcomes

The project is expected to provide insights into:

- Whether Bitcoin behaves more like a stock market asset or a safe-haven asset
- How stable or unstable correlations between Bitcoin and traditional assets are
- Whether Bitcoin can be considered a diversification tool in financial portfolios

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7. Tools and Technologies

The project will be implemented using the following tools:

- Python

- pandas, numpy
- matplotlib, seaborn
- yfinance
